Kernix 0.10.0 / Performance brief

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

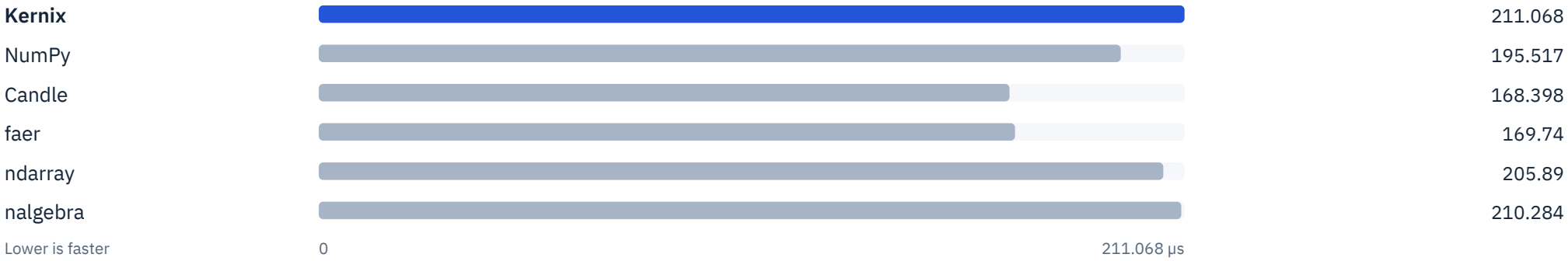
34 workloads

116 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.10.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



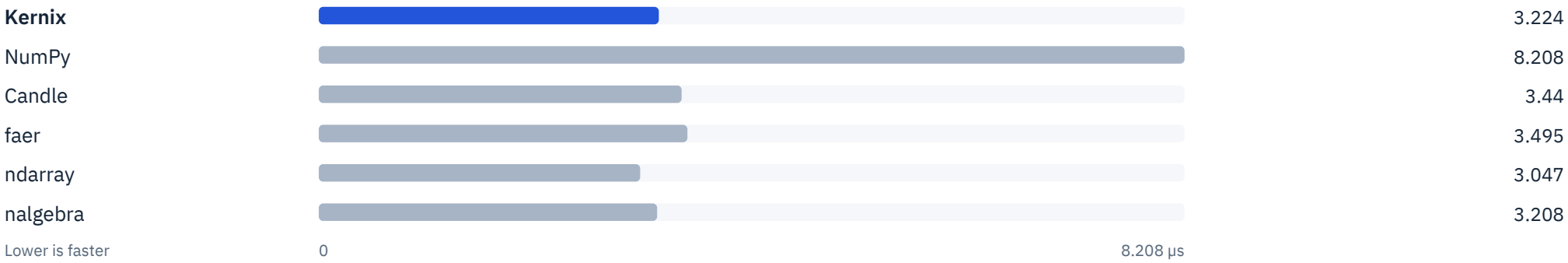
Kernix 0.10.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

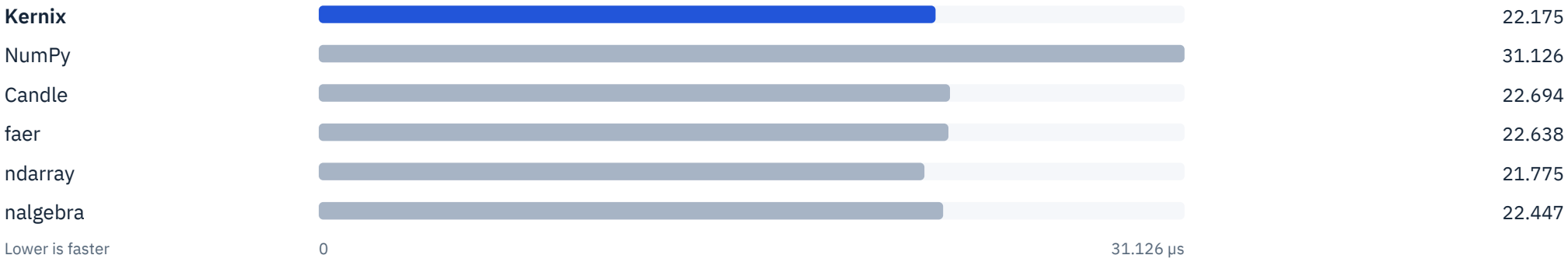
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



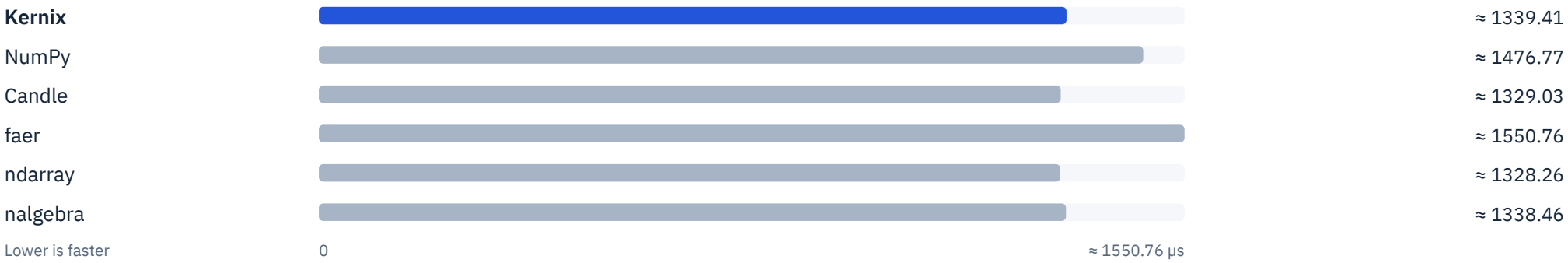
Kernix 0.10.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

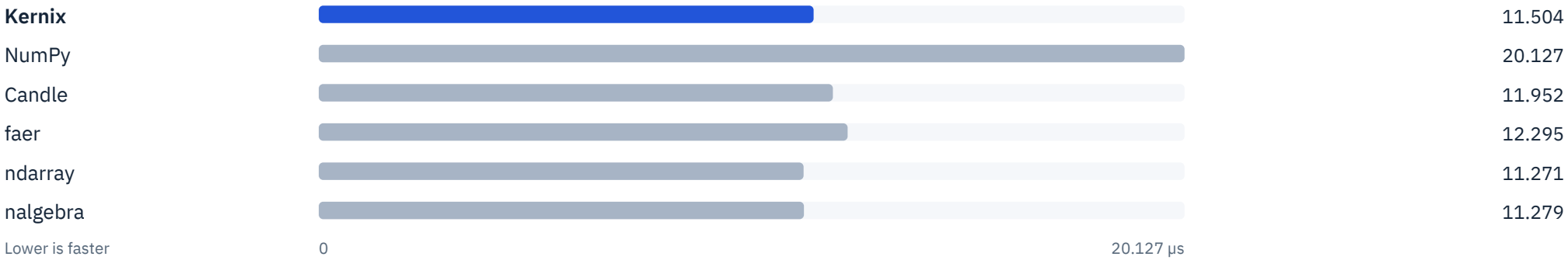
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



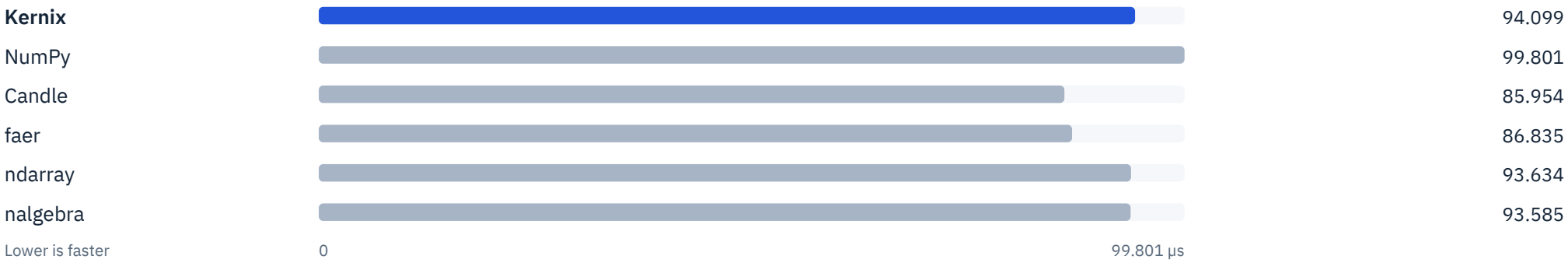
Kernix 0.10.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

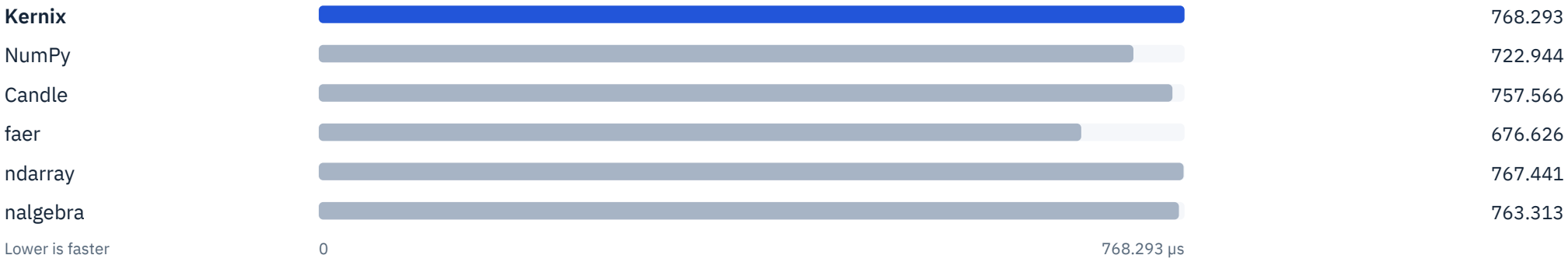
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.10.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.10.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.10.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



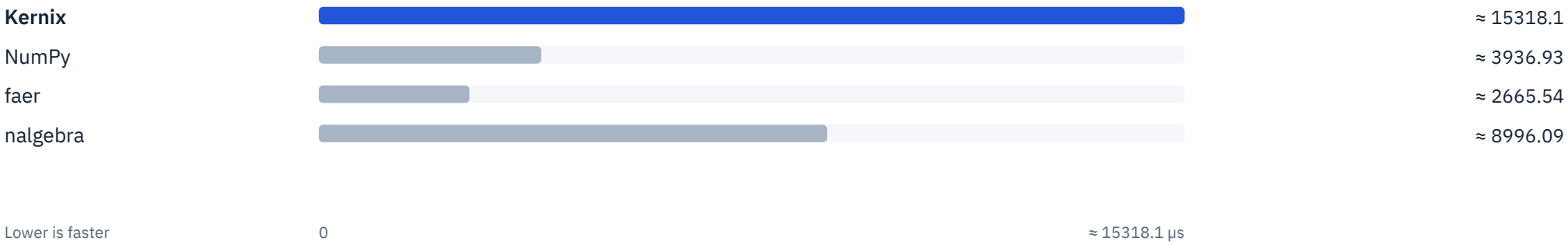
Kernix 0.10.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

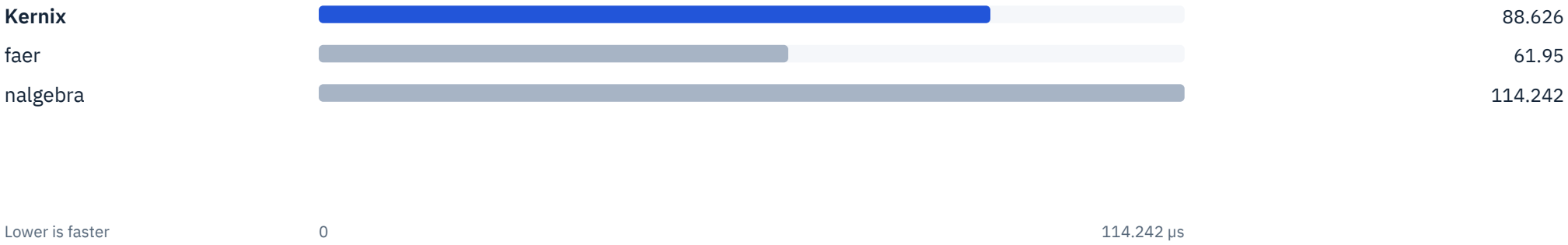
Per-call median



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



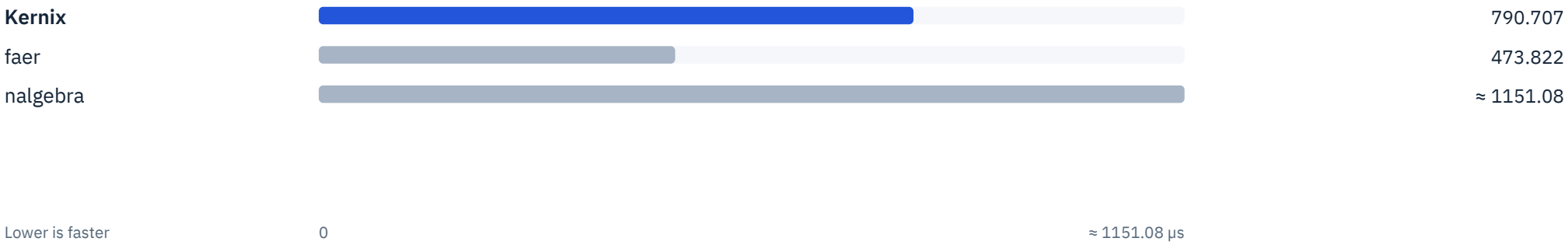
Kernix 0.10.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

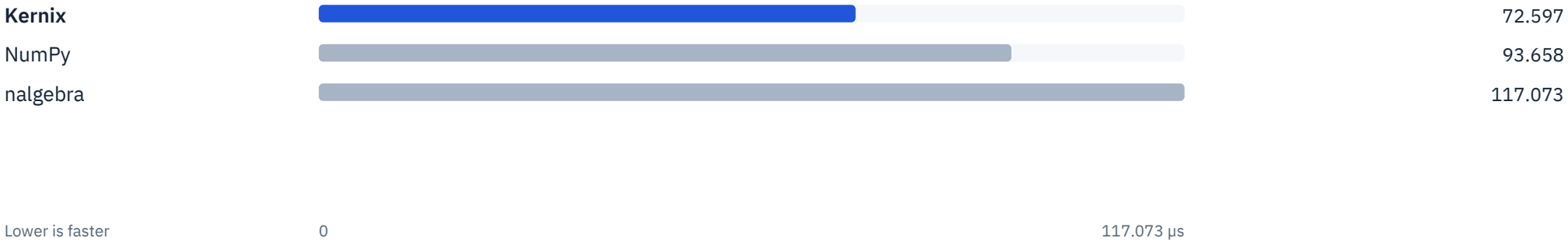
Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



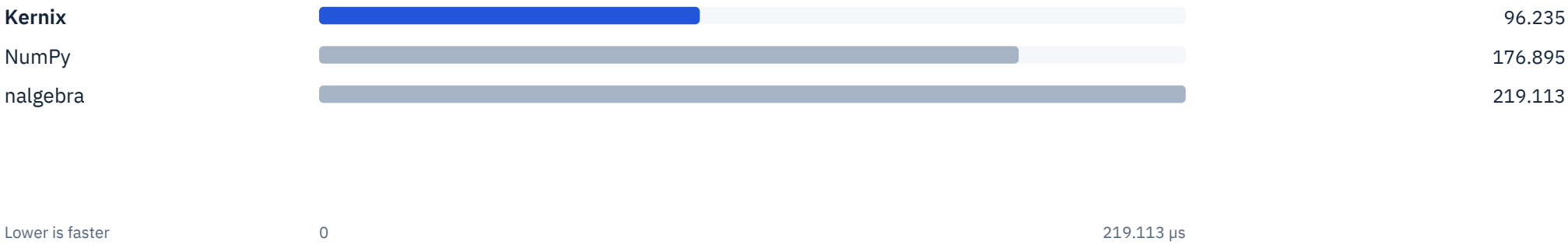
Kernix 0.10.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

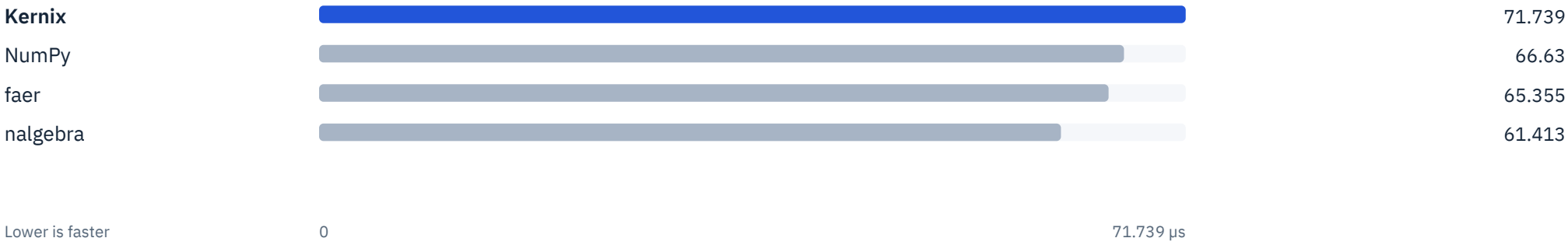
32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



Kernix 0.10.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

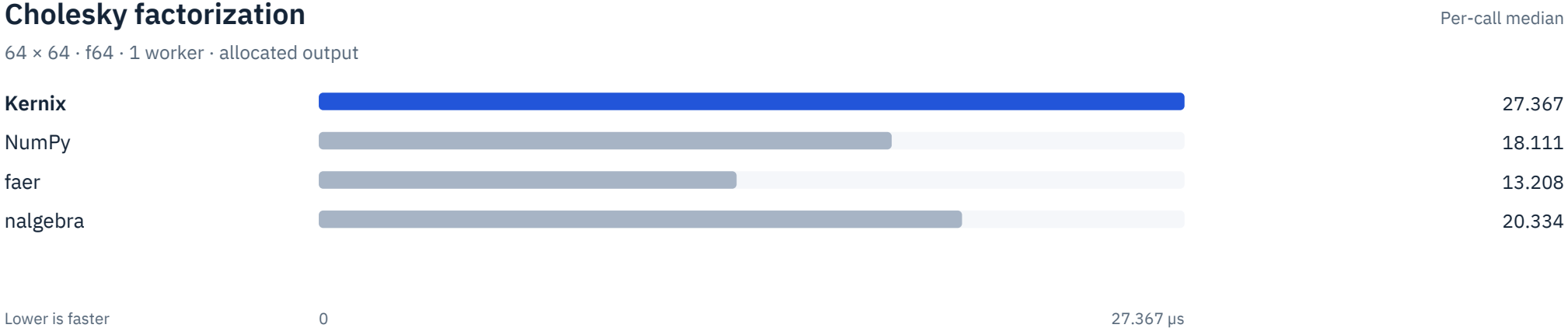
Thin QR

256 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

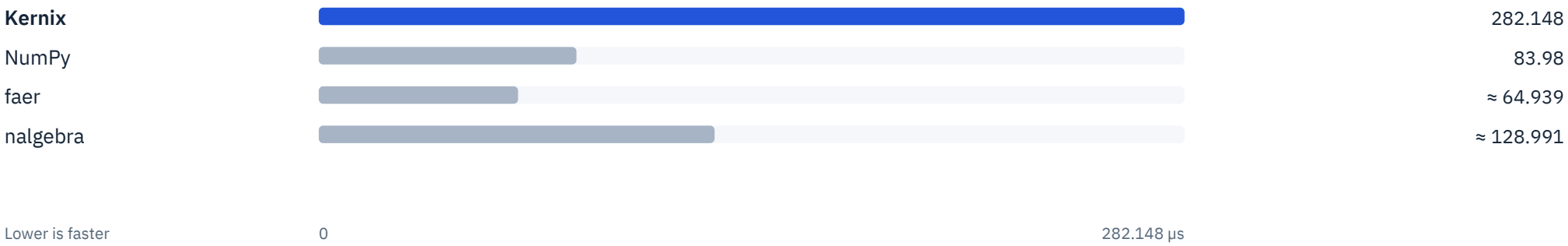


Kernix 0.10.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

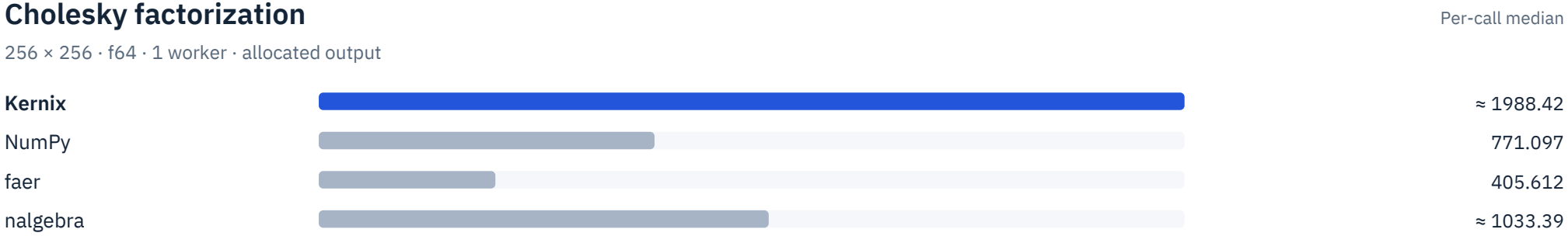
Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



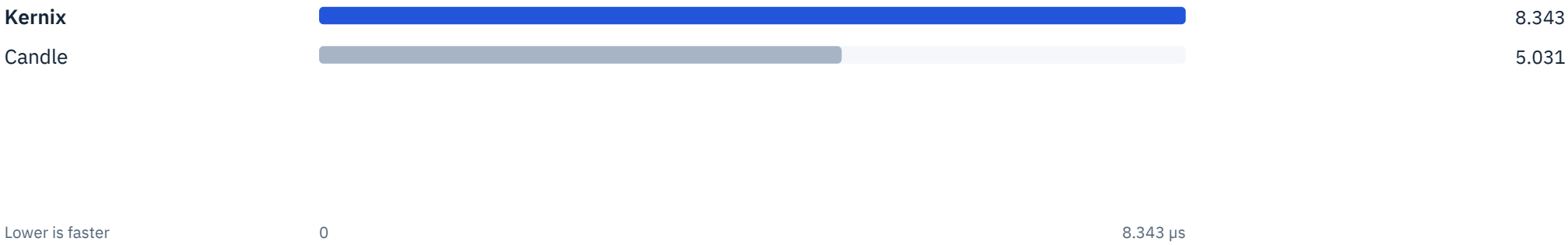
Kernix 0.10.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



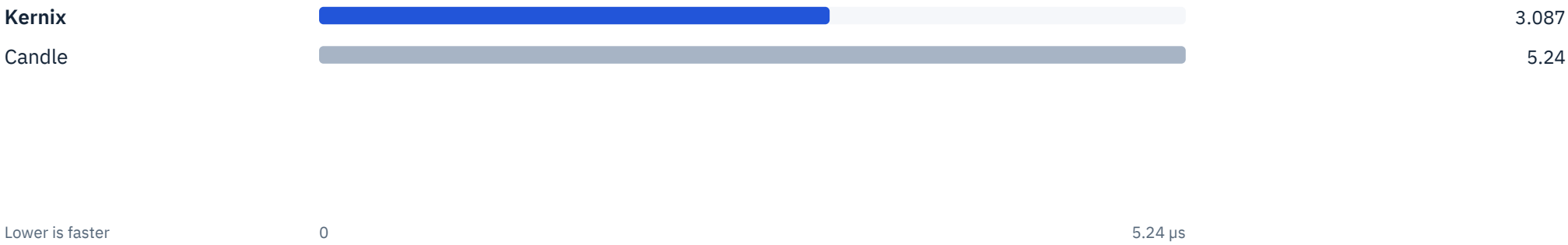
Kernix 0.10.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.10.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



Kernix 0.10.0 / Tensor contractions

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



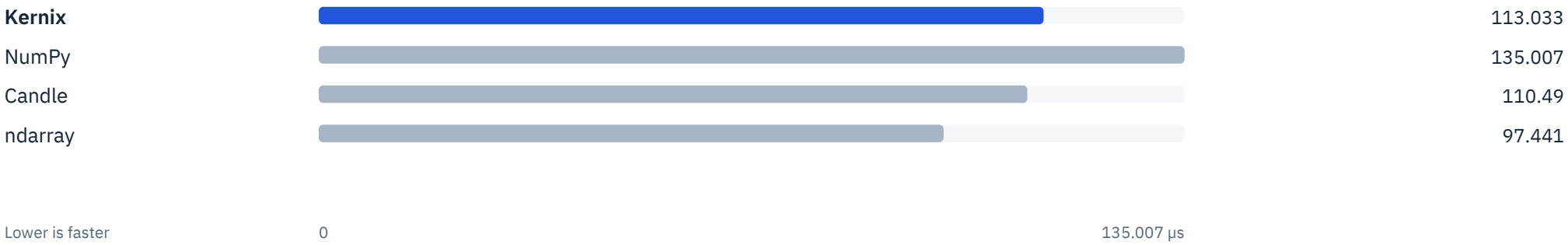
Kernix 0.10.0 / Statistics & activations

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Per-call median



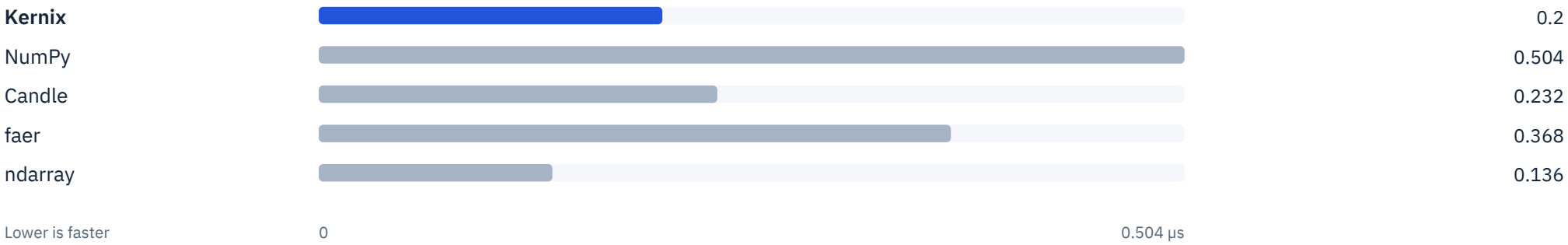
Kernix 0.10.0 / Tensor construction

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	211489	211.489
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	197028	197.028
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	100017	100.017
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	94587	94.587
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	3440	3.44
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	8208	8.208
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	3047	3.047
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	3208	3.208
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	3495	3.495
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	3224	3.224

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	21775	21.775
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	22694	22.694
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	22175	22.175
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	22447	22.447
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	31126	31.126
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	22638	22.638
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	195517	195.517
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	168398	168.398
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	210284	210.284
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	169740	169.74

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	205890	205.89
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	211068	211.068
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	1339411	1339.411
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	1476766	1476.766
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	1338464	1338.464
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1328265	1328.265
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	1329027	1329.027
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1550759	1550.759
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	11279	11.279
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	11952	11.952

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	12295	12.295
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	11271	11.271
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	11504	11.504
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	20127	20.127
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	94099	94.099
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	93634	93.634
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	85954	85.954
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	99801	99.801
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	93585	93.585
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	86835	86.835

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	768293	768.293
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	722944	722.944
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	767441	767.441
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	757566	757.566
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	676626	676.626
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	763313	763.313
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4439	4.439
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	12983	12.983
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3824	3.824
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	17174	17.174

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	97891	97.891
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	841096	841.096
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	49933	49.933
Cholesky factorization / nalgebra 512 × 512 · f64	8996093	8996.093
Cholesky factorization / NumPy 512 × 512 · f64	3936927	3936.927
Cholesky factorization / faer 512 × 512 · f64	2665544	2665.544
Cholesky factorization / Kernix 512 × 512 · f64	15318115	15318.115
Column-pivoted QR / Kernix 96 × 32 · f64	88626	88.626
Column-pivoted QR / faer 96 × 32 · f64	61950	61.95
Column-pivoted QR / nalgebra 96 × 32 · f64	114242	114.242

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 256 × 64 · f64	473822	473.822
Column-pivoted QR / nalgebra 256 × 64 · f64	1151077	1151.077
Column-pivoted QR / Kernix 256 × 64 · f64	790707	790.707
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	117073	117.073
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	93658	93.658
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	72597	72.597
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	219113	219.113
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	176895	176.895
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	96235	96.235
Thin QR / Kernix 96 × 32 · f64	71739	71.739

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 96 × 32 · f64	65355	65.355
Thin QR / NumPy 96 × 32 · f64	66630	66.63
Thin QR / nalgebra 96 × 32 · f64	61413	61.413
Thin QR / faer 256 × 64 · f64	447704	447.704
Thin QR / Kernix 256 × 64 · f64	694486	694.486
Thin QR / NumPy 256 × 64 · f64	573582	573.582
Thin QR / nalgebra 256 × 64 · f64	587694	587.694
Cholesky factorization / NumPy 64 × 64 · f64	18111	18.111
Cholesky factorization / nalgebra 64 × 64 · f64	20334	20.334
Cholesky factorization / faer 64 × 64 · f64	13208	13.208

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	27367	27.367
Cholesky factorization / Kernix 128 × 128 · f64	282148	282.148
Cholesky factorization / nalgebra 128 × 128 · f64	128991.00000000001	128.99100000000001
Cholesky factorization / faer 128 × 128 · f64	64938.99999999999	64.93899999999999
Cholesky factorization / NumPy 128 × 128 · f64	83980	83.98
Cholesky factorization / faer 256 × 256 · f64	405612	405.612
Cholesky factorization / Kernix 256 × 256 · f64	1988423	1988.423
Cholesky factorization / nalgebra 256 × 256 · f64	1033389	1033.389
Cholesky factorization / NumPy 256 × 256 · f64	771097	771.097
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	5031	5.031

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	8343	8.343
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3511	3.511
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7327	7.327
Model inference / Candle batched normalized/sliced MLP · f64	5240	5.24
Model inference / Kernix batched normalized/sliced MLP · f64	3087	3.087
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3839.9999999999995	3.8399999999999995
Model recording + forward / Candle batched normalized/sliced MLP · f64	7575	7.575
Model backward / Kernix prepared graph → fresh gradients · f64	4184	4.184
Model backward / Candle prepared graph → fresh gradients · f64	21135	21.135
Parameter update / Kernix transactional SGD, fixed gradients · f64	744	0.744

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	46693	46.693
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	16439	16.439
Row softmax / Candle 256 × 64 · f64	110490	110.49
Row softmax / ndarray 256 × 64 · f64	97441	97.441
Row softmax / Kernix 256 × 64 · f64	113033	113.033
Row softmax / NumPy 256 × 64 · f64	135007	135.007
Create zeros / Kernix 64 × 64 · f32	200	0.2
Create zeros / faer 64 × 64 · f32	368	0.368
Create zeros / ndarray 64 × 64 · f32	136	0.136
Create zeros / Candle 64 × 64 · f32	232	0.232

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	504	0.504
Create ones / Candle 64 × 64 · f32	344	0.344
Create ones / NumPy 64 × 64 · f32	1536	1.536
Create ones / Kernix 64 × 64 · f32	256	0.256
Create ones / ndarray 64 × 64 · f32	208	0.208
Create ones / faer 64 × 64 · f32	239.99999999999997	0.23999999999999997